MATH-18A *Exam 2 Review*

Made by Lena Pang

2.1 Polynomial Basics

- **degree** — highest exponent on variable
- **leading term** — term w/ highest exponent on variable; has the **leading coefficient** & **degree**

Things to Know
- Perfect Squares: $(ax+b)^2 = a^2x^2 + 2abx + b^2$
- Difference of Squares: $(ax+b)(ax-b) = a^2x^2 - b^2$

Techniques/Skills
- **adding/multiplying polynomials**: use distribution and then combine like terms

2.2 Factoring Polynomials

- **greatest common factor** — largest integer that divides every number
- **Zero Product Property** — if ab=0, then b=0 or b=0

Techniques/Skills
- **diamond method**: make & fill diamond; make & fill box; write out factors; solve for x with ZPP
- **quadratic formula**: set eq. to 0 → $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

2.3 Rational Equation

- **rational equation** — has variable in denominator
- **extraneous solution** — a solution that doesn't work in the original eq

Techniques/Skills
- **clearing fractions**: multiply by denominators to cancel out; simplify; check extran.; double-check

2.4 Quadratics & Parabolas

- **quadratic** — polynomial of degree 2
- **parabola** — the U-shape a quadratic makes
- **vertex** — maximum or minimum of the parabola
- **axis of symmetry** — splits a parabola into perfectly symmetrical halves (through vertex!)

Things to Know
- ABC-FORM: $y = ax^2 + bx + c$; y-intercept @ $(0,c)$
- VERTEX FORM: $y = a(x - h)^2 + k$; vertex @ (h,k)

Techniques/Skills
- **switch from ABC to vertex form**: completing the square, or just $h = -b/2a$ and $k = f(h)$ (plug in $x = h$)

2.5 Quadratic Word Problems

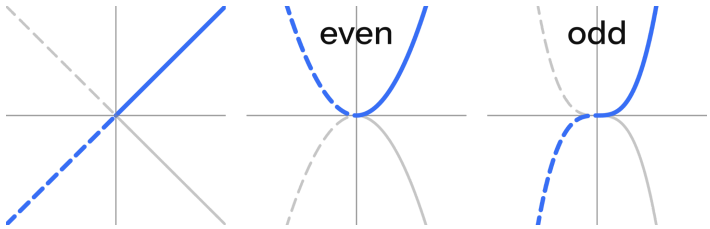
Things to Know
- SKETCH the problem!!!
- HEIGHT FORMULA: $H(t) = at^2 + bt + c$ (where a is always either -16 ft/s^2 or -4.5 m/s^2)

2.6 Dividing Polynomials

- **division algorithm format** — dividend = (divisor)(quotient) + remainder

Techniques/Skills
- **polynomial long division**: rewrite in descending order with 0's in missing terms; multiply to match first terms; subtract; repeat until degree of remainder < degree of divisor

2.7 Graphing Polynomials



- **real zero** — an x-intercept of a polynomial
- **multiplicity** — the exponent of a factor
- **end behavior** — whether the graph goes up or down at the ends (RHS and LHS)

Things to Know
- every real zero has a corresponding factor
- LEAD COEF positive: RHS ↗; negative: RHS ↘
- DEGREE odd: LHS matches; even: LHS opposite
- MULTIPLICITY even: bounces at real zero; odd: squiggles through real zero

Techniques/Skills
- **identifying real zeroes**: set each factor to 0; solve for x (example: 0 = $(x - 5)(x + 2)$ gives x = 5, -2)
- **equations → graphs**: identify real zeroes; find whether y-intercept is above (+) or below (-) x-axis; calculate leading term to determine end behavior; graph (beware of multiplicity!)
- **graphs → equations**: identify real zeroes; make a factor for each zero; add factor exponents for multiplicity as needed; put - in front if RHS ↘